

Statistical Methods for Computer Science

Exercises: Continuous Random Variables and Probability Distributions

2) Sales, investment, consumption, costs, and revenues are measures which can be represented by:

- A) discrete variable.
- B) fixed variables.
- C) continuous random variables.
- D) coefficients.

5) To obtain the probability that a continuous random variable falls in a specified range, we find the difference between the cumulative probabilities at the lower and upper end of the:

- A) range.
- B) median.
- C) mode.
- D) mean.

7) The function that may be used to find the probability that the random variable falls within a specific range is called the:

- A) expected value.
- B) uniform distribution.
- C) probability density function.
- D) linear function.

8) Because the probability of any specific value is 0 for a continuous random variable, the expected values for continuous random variables are computed using:

- A) Bayes' Theorem.
- B) an empirical formula.
- C) a Venn diagram.
- D) integral calculus.

11) The shape of the normal probability density function is a symmetric bell-shaped curve centered on the:

- A) variance.
- B) standard deviation.
- C) coefficient.
- D) mean.

12) We can convert any normal distribution to a distribution with a mean 0 and a variance 1. What is the distribution called?

- A) Poisson distribution
- B) standard normal distribution
- C) marginal distribution

15) Let the random variable Z follow a standard normal distribution. Find $P(0 < Z < 0.57)$.

- A) 0.2843
- B) 0.7843
- C) 0.2157
- D) 0.7157

16) Suppose that 19% of all sales are for amounts greater than \$1,000. In a random sample of 30 invoices, what is the probability that more than six of the invoices are for over \$1,000? Use the binomial approximation for the normal distribution.

- A) 0.4443
- B) 0.9440
- C) 0.5557
- D) 0.0560

18) Let the random variable Z follow a standard normal distribution. Find the value k , such that $P(Z > k) = 0.39$.

- A) 0.28
- B) 1.23
- C) -1.23
- D) -0.28

19) Let the random variable Z follow a standard normal distribution. Find $P(-2.21 < Z < 0)$.

- A) 0.9864
- B) 0.4864
- C) 0.0136
- D) 0.5136

21) Investment *A* has an expected return of 7.8% with a standard deviation of 2%. Investment *B* has an expected return of 7.2% with a standard deviation of 3.1%. Which stock is more likely to have a return greater than 10%?

- A) Stock *A*
- B) Stock *B*
- C) The probability is the same for both *A* and *B*.
- D) Unable to determine.

23) Let the random variable Z follow a standard normal distribution. Find $P(-1.33 < Z < 0.78)$.

- A) 0.6905
- B) 0.2823
- C) 0.2177
- D) 0.3095

THE NEXT QUESTIONS ARE BASED ON THE FOLLOWING INFORMATION:

You have recently joined a country club. The number of times you expect to play golf in a month is represented by a random variable with a mean of 10 and a standard deviation of 2.2. Assume you pay monthly membership fees of \$500 per month and pay an additional \$50 per round of golf.

24) What is your average monthly bill from the country club?

- A) \$700
- B) \$800
- C) \$900
- D) \$1000

25) What is the standard deviation for your average monthly bill from the country club?

- A) \$220
- B) \$110
- C) \$324
- D) \$180

26) Which of the following is false regarding the normal distribution?

- A) The mean, median, and mode are equal.
- B) The frequency of values peaks at the mean, regardless of the value of the mean or variance.
- C) 100% of the values fall between ± 3 standard deviations.
- D) The shape of the distribution is symmetrical around the mean.

27) One normal distribution has a mean of 5 and a standard deviation of 2. A second normal distribution has a mean of 6 and a standard deviation of 1. Which of the following statements is true?

- A) The width of the first distribution is wider than the second distribution.
- B) The central location of the first distribution is higher than the second distribution.
- C) The dispersions of both distributions are the same.
- D) The locations of both distributions are the same.

28) Let the random variable Z follow a standard normal distribution. Find the value k , such that $P(Z > k) = 0.73$.

- A) 0.27
- B) 0.73
- C) -0.16
- D) -0.61

THE NEXT QUESTIONS ARE BASED ON THE FOLLOWING INFORMATION:

Let the random variable X follow a normal distribution with a mean of 17.1 and a standard deviation of 3.2.

29) What is $P(X > 16)$?

- A) 0.3401
- B) 0.6331
- C) 0.3669
- D) 0.8326

30) What is $P(15 < X < 20)$?

- A) 0.5581
- B) 0.1814
- C) 0.5640
- D) 0.2546

31) The mean of a normally distributed group of weekly incomes for a group of executives is \$2,400 and the standard deviation is \$600. What is the z-score for an income of \$1,500/week?

- A) -1.5
- B) -1.0
- C) 2.5
- D) 1.9

32) Let the random variable Z follow a standard normal distribution. Find the value k , such that $P(-0.62 < Z < k) = 0.43$.

- A) 0.20
- B) 0.52
- C) 0.12
- D) 0.56

33) The normal probabilities are calculated using the table of standard normal distribution where the mean and standard deviations are, respectively:

- A) 1 and 1.
- B) 10 and 0.
- C) 0 and 1.
- D) 0 and 10.

THE NEXT QUESTIONS ARE BASED ON THE FOLLOWING INFORMATION:

Let the random variable Z follow a standard normal distribution.

34) What is $P(Z > 1.2)$?

- A) 0.1112
- B) 0.8849
- C) 0.1151
- D) 0.6112

35) What is $P(Z > -0.21)$?

- A) 0.4207
- B) 0.4168
- C) 0.5793
- D) 0.5832

36) What is $P(0.33 < Z < 0.45)$?

- A) 0.5443
- B) 0.0443
- C) 0.4557
- D) 0.1515

37) The distribution of annual incomes of a sample of college graduates is normally distributed with a mean of \$52,000 and a standard deviation of 1,000. About 68 percent of the incomes lie between what two income levels?

- A) 52,000 and 62,000
- B) 50,000 and 60,000
- C) 51,000 and 53,000
- D) 53,000 and 57,000

THE NEXT QUESTIONS ARE BASED ON THE FOLLOWING INFORMATION:

Let the random variable X follow a normal distribution with a mean of 61.7 and a standard deviation of 5.2.

38) What is the value of k such that $P(X > k) = 0.63$?

- A) 59.984
- B) 66.446
- C) 62.830
- D) 67.576

39) What is the value of k such that $P(59 < X < k) = 0.54$?

- A) 65.8
- B) 64.6
- C) 63.7
- D) 66.9

40) What proportion of the total area under the normal curve is within $\pm$ two standard deviations of the mean?

- A) About 50%
- B) About 68%
- C) About 95%
- D) About 99.7%

THE NEXT QUESTIONS ARE BASED ON THE FOLLOWING INFORMATION:

The number of orders that come into a mail-order sales office each month is normally distributed with a mean of 298 and a standard deviation of 15.4.

41) What is the probability that in a particular month the office receives more than 310 orders?

- A) 0.7823
- B) 0.2826
- C) 0.7174
- D) 0.2177

42) The probability is 0.3 that the sales office receives less than how many orders?

- A) 310.9
- B) 285.1
- C) 290.0
- D) 306.0

43) What is the area under the normal curve between $Z = 1.0$ and $Z = 2.0$?

- A) 0.0228
- B) 0.1359
- C) 0.3413
- D) 0.4772

44) What is the area under the normal curve between $Z = -2$ and $Z = 0$?

- A) 0.5000
- B) 0.6826
- C) 0.1359
- D) 0.4772

45) What proportion of the total area under the normal curve is above the mean?

- A) 50%
- B) 68%
- C) 95%
- D) 99.7%

50) The median score for the final exam in a chemistry class was 75 and 18% of the students received a grade greater than 80. If the test scores were normally distributed, what is the probability that a randomly selected exam had a score between 75 and 80?

- A) 68%
- B) 32%
- C) 80%
- D) 50%

116) Many measures such as length, weight, sales, investment, consumption, costs, and revenues can be represented by continuous random variables.

118) The probability that a random variable lies between a pair of values is the area under the probability density function to the left of the two values.

119) The mean of a probability distribution provides a measure of the center of the distribution.

120) The variance of a probability distribution provides a measure of the average spread of a distribution.

122) The total area under the curve of the probability density function for a continuous random variable depends on the shape of the probability distribution of the random variable.

123) In any normal distribution, the mean, median, mode, and standard deviation are all at the same position on the horizontal axis.

124) Any set of normally distributed data can be transformed to a standardized form.

125) In the normal distribution, the curve approaches, but never intercepts the horizontal axis either to the left or right.

126) The shape of the normal probability distribution is determined by the population standard deviation σ . Small values of σ reduce the height of the curve and increase the spread; large values of σ increase the height of the curve and reduce the spread.

127) The standard normal distribution is the most well-known discrete distribution in statistics.

128) A continuous random variable X is normally distributed with a mean of 1200 and a standard deviation of 150. Given that $X = 1410$, its corresponding Z -score is 1.40.

130) In the normal distribution, the total area beneath the curve represents the probability for all possible outcomes for a given event.

131) If the random variable Z follows a standard normal distribution, then, $\sigma_Z^2 = \sigma_z^2$.

132) If the random variable $Y = a + bX$, then $\sigma_y^2 = b^2 \sigma_x^2$.

133) The total area under the curve of the probability density function of any normal distribution is always 1.0.

134) If the random variable Z follows a standard normal distribution, then $\sigma_z^2 = 1$.

135) The normal curve gets closer and closer to the X -axis but never actually touches it.

137) There is not just one normal probability distribution; there is a family of distributions.

138) If X is a normal random variable with mean $\mu = 4$ and standard deviation $\sigma = 2$, and Y is a normal random variable with mean $\mu = 10$ and standard deviation $\sigma = 5$, then the probabilities $P(X < 0)$ and $P(Y < 0)$ are equal.

140) If X is a normal random variable with mean of 2 and standard deviation of 5, then $P(X < 3) = P(X > 7)$.

167) Investment A has an expected return of 10% with a standard deviation of 3.5%. Investment B has an expected return of 6% with a standard deviation of 1.2%. If you invest equally in both investments, what is the expected return and standard deviation of your portfolio? What assumptions have you made?

THE NEXT QUESTIONS ARE BASED ON THE FOLLOWING INFORMATION:

Let the random variable Z follow a standard normal distribution.

176) What is $P(Z > 0.29)$?

177) What is $P(Z < 1.23)$?

178) What is $P(Z > -0.52)$?

179) What is $P(-0.44 < Z < 1.2)$?

180) As manager of a pizza shop, you are responsible for placing the food orders. You currently have enough anchovies for 8 pizzas. You expect to have orders for 60 pizzas tonight. If 8% of all pizzas are ordered with anchovies, what is the probability that you run out of anchovies before the evening is over? Use the normal approximation for the binomial.

THE NEXT QUESTIONS ARE BASED ON THE FOLLOWING INFORMATION:

Let the random variable Z follow a standard normal distribution.

181) Find the value k such that $P(Z > k) = 0.83$

182) Find the value k such that $P(Z > k) = 0.43$

183) Find the value k such that $P(0 < Z < k) = 0.35$

184) Find the value k such that $P(-0.71 < Z < k) \approx 0.67$

185) The time it takes to assemble an electronic component is normally distributed with a mean of 17.2 minutes and a standard deviation of 3.1 minutes. The probability is 90% that it will take at least how long to assemble a component?

188) In a recent survey of high school students, it was found that the average amount of money spent on entertainment each week was normally distributed with a mean of \$52.30. Suppose you are told that there is an 80% probability that a randomly-selected student spends somewhere between \$49.74 and \$54.86. What is the standard deviation of the amount of money spent by high school students monthly?

194) Sales at a local plumbing wholesaler consist of both over-the-counter sales as well as deliveries. During the course of a month, over-the-counter sales have a mean of \$102,972 with a standard deviation of \$13,523. Over the same time period, deliveries average \$242,354 with a standard deviation of \$24,956. Assuming that the sales over-the-counter are independent of deliveries, what are the mean and standard deviation of the wholesaler monthly sales?

196) The number of viewers ordering a particular pay-per-view program is normally distributed. 20% of the time, fewer than 20,000 people order the program. Only ten percent of the time more than 28,000 people order the program. What is the mean and standard deviation of the number of people ordering the program?